

CSA1711 - Artificial Intelligence

CO3 Assessment Tool - 1

Case Study Assignment - Answer Script

Case Study 1: Smart Medical Diagnosis System

Scenario: A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis using Propositional Logic rules relating Fever, Cough, Rash, and Breathlessness to possible Flu, Measles, and Pneumonia. Patient A presents with Fever = True, Cough = True, Breathlessness = True.

(a) Propositional Logic Representation

Symbol definitions:

- F = 'Patient has Fever'
- C = 'Patient has Cough'
- R = 'Patient has Rash'
- B = 'Patient has Breathlessness'
- Flu = 'Patient possibly has Flu'
- Measles = 'Patient possibly has Measles'
- Pneumonia = 'Patient possibly has Pneumonia'

Rules in propositional logic:

Rule 1: $F \wedge C \rightarrow \text{Flu}$

Rule 2: $F \wedge R \rightarrow \text{Measles}$

Rule 3: $C \wedge B \rightarrow \text{Pneumonia}$

Facts about Patient A:

F = True, C = True, B = True (R is not asserted/unknown)

(b) Forward Chaining

Forward chaining starts from the known facts and fires any rule whose premises are fully satisfied, adding new derived facts to the knowledge base, repeating until no new fact can be derived.

Step	Known Facts	Rule Checked	Fired?	Fact Derived
1	F, C, B	Rule 1: $F \wedge C \rightarrow \text{Flu}$	Yes (F=T, C=T)	Flu
2	F, C, B, Flu	Rule 2: $F \wedge R \rightarrow \text{Measles}$	No (R is unknown)	— none —
3	F, C, B, Flu	Rule 3: $C \wedge B \rightarrow \text{Pneumonia}$	Yes (C=T, B=T)	Pneumonia
4	F, C, B, Flu, Pneumonia	All rules re-checked	No further rule fires	— none —

Final derived diagnoses for Patient A: Possible Flu and Possible Pneumonia (Measles cannot be derived since Rash is not confirmed).

(c) Backward Chaining — Goal: 'Patient A has Pneumonia'

Backward chaining starts from the goal and recursively reduces it to sub-goals until each sub-goal matches a known fact.

Goal reduction tree:

Pneumonia

└─ requires Rule 3: $C \wedge B \rightarrow \text{Pneumonia}$

 └─ Sub-goal 1: C (Cough) → matches fact 'Cough = True' ✓

└ Sub-goal 2: B (Breathlessness) → matches fact 'Breathlessness = True' ✓
 Both sub-goals satisfied ⇒ Rule 3 fires ⇒ Pneumonia is CONFIRMED TRUE for Patient A.

Case Study 2: Autonomous Traffic Management Agent

Scenario: A smart city traffic controller uses an FOL-based knowledge base with rules for clearing paths, setting signals, and allowing following vehicles to proceed for emergency vehicle routing, with facts about an Ambulance and a following CarA.

(a) Unification of Rule 1 with the Given Facts

Rule 1: $\forall x: \text{Vehicle}(x) \wedge \text{EmergencyType}(x) \rightarrow \text{ClearPath}(x)$

Facts: $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$

Step 1: Unify $\text{Vehicle}(x)$ with $\text{Vehicle}(\text{Ambulance})$

$\theta_1 = \{ x / \text{Ambulance} \}$

Step 2: Unify $\text{EmergencyType}(x)$ with $\text{EmergencyType}(\text{Ambulance})$ using θ_1

$\text{EmergencyType}(\text{Ambulance})$ already matches $\text{EmergencyType}(\text{Ambulance})$ under θ_1 – consistent, no conflict.

Final MGU: $\theta = \{ x / \text{Ambulance} \}$

Applying θ to the rule's conclusion gives the new derived fact: $\text{ClearPath}(\text{Ambulance})$.

(b) Resolution Proof that 'Proceed(CarA)' is True

Step 1 — Convert all relevant rules to Conjunctive Normal Form (CNF):

Rule 1 (CNF): $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$

Rule 2a (CNF): $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$

Rule 2b (CNF): $\neg \text{ClearPath}(x) \vee \neg \text{RedSignal}(x)$

Rule 3 (CNF): $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$

Facts (unit clauses): $\text{Vehicle}(\text{Ambulance})$, $\text{EmergencyType}(\text{Ambulance})$, $\text{Vehicle}(\text{CarA})$, $\text{Behind}(\text{CarA}, \text{Ambulance})$

Negated goal (for refutation): $\neg \text{Proceed}(\text{CarA})$

Step 2 — Resolution steps:

#	Resolve	Substitution	Resulting Clause
1	$\text{Vehicle}(\text{Ambulance})$ with $\neg \text{Vehicle}(x) \vee \neg \text{EmergencyType}(x) \vee \text{ClearPath}(x)$	$x / \text{Ambulance}$	$\neg \text{EmergencyType}(\text{Ambulance}) \vee \text{ClearPath}(\text{Ambulance})$
2	$\text{EmergencyType}(\text{Ambulance})$ with result of step 1	—	$\text{ClearPath}(\text{Ambulance})$
3	$\text{ClearPath}(\text{Ambulance})$ with $\neg \text{ClearPath}(x) \vee \text{GreenSignal}(x)$	$x / \text{Ambulance}$	$\text{GreenSignal}(\text{Ambulance})$
4	$\text{Vehicle}(\text{CarA})$ with $\neg \text{Vehicle}(x) \vee \neg \text{Behind}(x, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(x)$	x / CarA	$\neg \text{Behind}(\text{CarA}, y) \vee \neg \text{GreenSignal}(y) \vee \text{Proceed}(\text{CarA})$
5	$\text{Behind}(\text{CarA}, \text{Ambulance})$ with result of step 4	$y / \text{Ambulance}$	$\neg \text{GreenSignal}(\text{Ambulance}) \vee \text{Proceed}(\text{CarA})$
6	$\text{GreenSignal}(\text{Ambulance})$ (from step 3) with result of step 5	—	$\text{Proceed}(\text{CarA})$
7	$\text{Proceed}(\text{CarA})$ with $\neg \text{Proceed}(\text{CarA})$ (negated goal)	—	Empty clause $\square$

Empty clause $\square$ derived ⇒ contradiction ⇒ the negated goal is false ⇒ $\text{Proceed}(\text{CarA})$ is PROVEN TRUE.

(c) Is the FOL Knowledge Base Sufficient for Two Simultaneous Emergency Vehicles?

No, the current knowledge base is not sufficient. All three rules operate on a single variable x independently, with no predicate capturing conflicts between two emergency vehicles approaching the same intersection from different directions. If two emergency vehicles x and y both satisfy $\text{Vehicle} \wedge \text{EmergencyType}$, Rule 1 and Rule 2 would independently derive $\text{ClearPath}(x)$, $\text{ClearPath}(y)$, $\text{GreenSignal}(x)$, and $\text{GreenSignal}(y)$ simultaneously — which is logically consistent within the KB but physically unsafe if x and y approach from conflicting directions at the same junction.

Additional rules/facts needed:

- A $\text{Conflict}(x,y)$ predicate identifying pairs of vehicles/directions that cannot both receive a green signal at the same time.
- A $\text{Priority}(x, \text{level})$ fact assigning relative priority among emergency vehicle types (e.g., Ambulance vs Fire Truck vs Police).
- A conflict-resolution rule such as: $\forall x \forall y: \text{EmergencyType}(x) \wedge \text{EmergencyType}(y) \wedge \text{Conflict}(x,y) \wedge \text{HigherPriority}(x,y) \rightarrow \text{GreenSignal}(x) \wedge \neg \text{GreenSignal}(y)$, to prevent the KB from deriving contradictory/unsafe simultaneous green signals.

Justification: monotonic forward/backward chaining will happily derive all logically entailed conclusions from the given rules, even if two derived conclusions are jointly unsafe in the real world. Since safety here depends on relative priority and spatial conflict — concepts the current KB has no vocabulary for — the KB must be extended with explicit conflict and priority predicates before it can correctly reason about two simultaneous emergency vehicles.

Case Study 3: Agricultural Expert Reasoning System

Scenario: An agricultural advisory logical agent reasons with propositional rules relating soil condition and crop type to irrigation and risk decisions, with facts $\text{SoilDry} = \text{True}$ and $\text{CropWheat} = \text{True}$.

(a) CNF Conversion of P1, P2, P3 and the Facts

P1: $\text{SoilDry} \rightarrow \text{IrrigationNeeded}$

Eliminate implication: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

No negation to move inward, no distribution needed.

CNF: $\neg \text{SoilDry} \vee \text{IrrigationNeeded}$

P2: $\text{IrrigationNeeded} \wedge \text{CropWheat} \rightarrow \text{ApplyDripMethod}$

Eliminate implication: $\neg(\text{IrrigationNeeded} \wedge \text{CropWheat}) \vee \text{ApplyDripMethod}$

Move $\neg$ inward (De Morgan): $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

CNF: $\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$

P3: $\neg \text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$

Eliminate implication: $\neg(\neg \text{ApplyDripMethod} \wedge \text{CropWheat}) \vee \text{CropAtRisk}$

Move $\neg$ inward (De Morgan): $(\text{ApplyDripMethod} \vee \neg \text{CropWheat}) \vee \text{CropAtRisk}$

Distribute (already a flat disjunction, no further distribution needed):

CNF: $\text{ApplyDripMethod} \vee \neg \text{CropWheat} \vee \text{CropAtRisk}$

Facts as unit clauses:

SoilDry

CropWheat

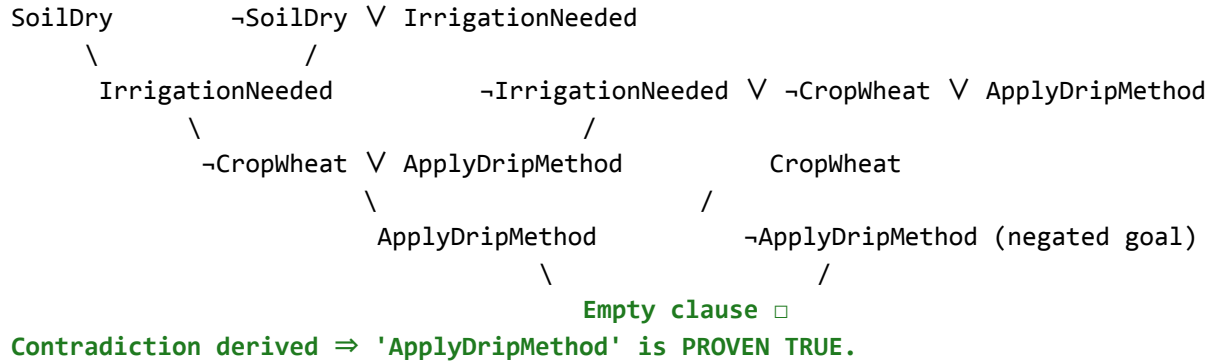
(b) Resolution Refutation — Prove 'ApplyDripMethod'

Negate the goal and add it to the CNF knowledge base: $\neg \text{ApplyDripMethod}$

#	Clause A	Clause B	Resolvent
1	SoilDry	$\neg \text{SoilDry} \vee \text{IrrigationNeeded}$	IrrigationNeeded
2	IrrigationNeeded (step 1)	$\neg \text{IrrigationNeeded} \vee \neg \text{CropWheat} \vee \text{ApplyDripMethod}$	$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$
3	CropWheat	$\neg \text{CropWheat} \vee \text{ApplyDripMethod}$ (step 2)	ApplyDripMethod

#	Clause A	Clause B	Resolvent
4	ApplyDripMethod (step 3)	$\neg$ ApplyDripMethod (negated goal)	Empty clause $\square$

Resolution proof tree:



(c) Effect of Removing the Fact 'SoilDry'

Step-by-step effect: without the unit clause SoilDry, resolution step 1 above can no longer cancel the $\neg$ SoilDry literal in P1's clause, so IrrigationNeeded can no longer be derived. Since IrrigationNeeded is unavailable, step 2 cannot fire either, so ApplyDripMethod also cannot be derived. The resolution refutation for 'ApplyDripMethod' therefore fails to reach the empty clause — the theorem is no longer provable from the KB.

Whether 'CropAtRisk' still follows depends on the reasoning pattern applied:

- Under strict classical (monotonic) resolution: $\neg$ ApplyDripMethod is never explicitly proven — only 'ApplyDripMethod is not entailed' — so P3's clause ($\text{ApplyDripMethod} \vee \neg\text{CropWheat} \vee \text{CropAtRisk}$) cannot resolve away its ApplyDripMethod literal. CropAtRisk therefore does NOT logically follow; the KB is simply silent about it (the query 'CropAtRisk' is undetermined).
- Under a Closed-World Assumption / negation-as-failure (common in rule-based expert systems): since ApplyDripMethod cannot be proven, it is treated as false by default. Combined with the fact CropWheat = True, Rule P3 ($\neg\text{ApplyDripMethod} \wedge \text{CropWheat} \rightarrow \text{CropAtRisk}$) then fires, and CropAtRisk IS concluded True.

Conclusion: the answer is reasoning-pattern-dependent — a strict, sound resolution-refutation engine cannot prove CropAtRisk once SoilDry is removed, whereas a practical rule-based expert system using default/closed-world reasoning would still flag CropAtRisk as True, since it cannot confirm ApplyDripMethod was carried out.

Case Study 4: Wumpus World Knowledge Agent

Scenario: An AI agent navigating the Wumpus World perceives Stench at [1,2], Breeze at [1,1], and Glitter at [2,2], and reasons using propositional rules relating stench/breeze/glitter percepts to Wumpus, Pit, Gold, and Safe predicates.

(a) Belief State Construction via Modus Ponens

Applying Modus Ponens (from P and $P \rightarrow Q$, infer Q) to each percept and rule:

Percept (Fact)	Rule Applied	Conclusion Derived (Modus Ponens)
Stench[1,2] = True	Rule 1: Stench[1,2] $\rightarrow$ Wumpus adjacent to [1,2]	Wumpus is adjacent to [1,2] $\Rightarrow$ Wumpus $\in \{[1,1], [1,3], [2,2]\}$
Breeze[1,1] = True	Rule 2: Breeze[1,1] $\rightarrow$ Pit adjacent to [1,1]	Pit is adjacent to [1,1] $\Rightarrow$ Pit $\in \{[1,2], [2,1]\}$
Glitter[2,2] = True	Rule 3: Glitter[x,y] $\rightarrow$ Gold at [x,y]	Gold is at [2,2]

Refinement using survival evidence: since the agent physically stood at [1,1] (start), at [1,2] (to perceive the stench), and at [2,2] (to perceive the glitter) and survived at each, none of these three cells can contain the Wumpus or a Pit. Applying Rule 4 ($\neg\text{Wumpus}[x,y] \wedge \neg\text{Pit}[x,y] \rightarrow \text{Safe}[x,y]$) to each of these cells:

- [1,1]: $\neg\text{Wumpus}[1,1] \wedge \neg\text{Pit}[1,1] \rightarrow \text{Safe}[1,1]$
- [1,2]: $\neg\text{Wumpus}[1,2] \wedge \neg\text{Pit}[1,2] \rightarrow \text{Safe}[1,2]$
- [2,2]: $\neg\text{Wumpus}[2,2] \wedge \neg\text{Pit}[2,2] \rightarrow \text{Safe}[2,2]$

List of all derived conclusions:

- Wumpus $\in \{[1,3], [2,2]\}$ (after eliminating $[1,1]$ as a candidate, since $[1,1]$ is confirmed safe)
- Pit $\in \{[1,2], [2,1]\}$ (candidate set from Rule 2)
- Gold is at $[2,2]$
- Safe $[1,1]$, Safe $[1,2]$, Safe $[2,2]$ all confirmed

(b) Forward Chaining — Locating the Wumpus and the Pit

Step	Reasoning	Result
1	Stench $[1,2]$ fires Rule 1 $\rightarrow$ Wumpus adjacent to $[1,2]$	Candidate set: $\{[1,1], [1,3], [2,2]\}$
2	$[1,1]$ already confirmed Safe (agent's start cell) $\rightarrow$ eliminate $[1,1]$ from candidates	Wumpus $\in \{[1,3], [2,2]\}$
3	Agent stood at $[2,2]$ and survived (perceived Glitter there) $\rightarrow [2,2]$ confirmed Safe $\rightarrow$ eliminate $[2,2]$	Wumpus $\in \{[1,3]\} \Rightarrow$ Wumpus is at $[1,3]$
4	Breeze $[1,1]$ fires Rule 2 $\rightarrow$ Pit adjacent to $[1,1]$	Candidate set: $\{[1,2], [2,1]\}$
5	Agent stood at $[1,2]$ and survived (perceived Stench there) $\rightarrow [1,2]$ confirmed Safe $\rightarrow$ eliminate $[1,2]$	Pit $\in \{[2,1]\} \Rightarrow$ Pit is at $[2,1]$

Conclusion: The Wumpus is located at $[1,3]$ and the Pit is located at $[2,1]$.

(c) Safety Evaluation of Path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$

Cell on Path	Status (from parts a & b)	Contains Wumpus?	Contains Pit?
$[1,1]$	Safe (start cell, confirmed via Rule 4)	No	No
$[1,2]$	Safe (agent survived there, confirmed via Rule 4)	No	No
$[2,2]$	Safe (agent survived there; Gold located here)	No	No

Since the Wumpus is located at $[1,3]$ and the Pit is located at $[2,1]$ — neither of which lies on the path $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ — and each cell on this path has been independently confirmed Safe by Modus Ponens applied to Rule 4 using the percept evidence from parts (a) and (b), the path is logically proven to be entirely safe.

Conclusion: The agent CAN safely traverse $[1,1] \rightarrow [1,2] \rightarrow [2,2]$ to collect the Gold, with no risk of encountering the Wumpus or falling into the Pit.